Fiscal Year:	FY 2006	**Task Last Updated:**	FY 03/15/2006
PI Name:	Braby, Leslie Ph.D.		
Project Title:	Methods for Real Time Measurement of Dose and Charged Particle Spectrum		
Division Name:	Human Research		
Program/Discipline:	HUMAN RESEARCH		
Program/Discipline--Element/Subdiscipline:			
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **SR**:Space Radiation		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	labraby@tamu.edu	**Fax:**	FY
PI Organization Type:	UNIVERSITY	**Phone:**	979.862.1798
Organization Name:	Texas Engineering Experiment Station		
PI Address 1:	3133 TAMU		
PI Address 2:			
PI Web Page:			
City:	College Station	**State:**	TX
Zip Code:	77843	**Congressional District:**	17
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2003 Space Radiation Biology and Shielding 03-OBPR-07
Start Date:	05/15/2004	**End Date:**	03/31/2009
No. of Post Docs:	0	**No. of PhD Degrees:**	0
No. of PhD Candidates:	2	**No. of Master' Degrees:**	0
No. of Master's Candidates:	1	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA JSC
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:	NOTE: Received NCE to 3/31/2009 per PI (1/10) NOTE: Received NCE to 9/1/2007 (from previous end date of 6/1/2007) per PI (6/07)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):			
Grant/Contract No.:	NNJ04HH02G		
Performance Goal No.:			
Performance Goal Text:			
Task Description:	Low pressure, tissue equivalent, proportional counters are routinely used to measure the dose and radiation quality in space. A new approach to colleting and analyzing data will be developed in order to provide the information needed to improved models of cancer risk and to better evaluate radiation shielding. By measuring lineal energy simultaneously in 3 different size sites we expect to be able to distinguish between high energy heavy ions and lower energy lighter ions that have the same stopping power.		

Rationale for HRP Directed Research:

Research Impact/Earth Benefits:

Task Progress:

Low pressure, tissue equivalent, proportional counters are routinely used to measure the dose and radiation quality in space. The techniques and nomenclature of microdosimetry are generally used to describe the results of these measurements. The lineal energy distribution and the number of events, both obtained from a single tissue equivalent proportional counter measurement, provide an accurate measure of the dose and an estimate of the radiation quality for mixed radiation fields, such as those found in space. However, the same dose and lineal energy distribution can be produced by different combinations of cosmic ray particles and energies. Differences in the fluence spectrum may be responsible for differences in the biological effectiveness of particles with the same LET (linear energy transfer), so additional information about the radiation should improve predictions of the consequences of radiation exposure. We believe that an advanced approach to collecting and analyzing tissue equivalent proportional counter data can provide the information needed for improved models of cancer risk and to better evaluate radiation shielding. Information about delta ray range, and, therefore, primary ion velocity, can be obtained by making measurements in simulated sites of different sizes. For a given high energy heavy ion, the mean lineal energy decreases with decreasing site size because a larger fraction of the delta ray energy is deposited outside the site. By measuring lineal energy simultaneously in 3 different size sites, we expect to be able to distinguish between high energy heavy ions and lower energy lighter ions that have the same stopping power.

To test this approach GEANT4 Monte Carlo calculations for walled detectors and for sites in a continuous homogeneous medium are being pursued. The initial estimates of energy deposition that showed the potential usefulness of this approach were for defined sites in an infinite homogeneous medium. Detectors that can simulate this condition, referred to as wall-less proportional counters, can be built, but they are inherently larger and more complex than the conventional solid walled proportional counters currently used in dosimetry systems. However, solid walled detectors are expected to be less sensitive to delta ray escape because the physical scale of the curvature and spacing of delta ray tracks is compressed by approximately a factor of 105 in the solid wall relative to that in the collection volume of a low pressure proportional counter, and the probability of a delta ray reentering a specified site is much higher if the site is surrounded by a solid wall. This “wall effect” does not alter the dose measured by a proportional counter, but it does reduce the difference between the stopping power of the ion and the mean lineal energy measured in a small site because delta rays which would occur as independent events in a uniform medium occur in coincidence in the solid walled detector.

Simulation of energy deposition in a specified site in a homogeneous medium irradiated by high velocity particles is challenging. In principal, the beam diameter should be twice the maximum range of the delta rays produced by the incident radiation, generally thousands of times the site diameter. The probability of a delta ray produced by a primary ion near the edge of the beam interacting with the site at the center of the beam is extremely small due to solid angle considerations. This makes simulation using a homogeneous beam entirely impractical in terms of computing time. However, the problem can be solved by dividing it into two parts. A uniform beam is used in the simulation for small impact parameters and the results of this calculation are combined with appropriately weighted spectra for delta rays calculated at larger impact parameters.

In the case of the solid walled detector the calculation is faster because the solid angle of the detector volume is relatively large for most primary tracks within the range of the delta rays. In this case, the frequency mean specific energy, yF, decreases, as expected, as the site size decreases. However, the magnitude of the decrease is more than was anticipated, considering that the wall effect tends to increase yF and that the results of measurements made with 2 micrometer diameter detectors on the space station have mean values only slightly below the expected mean LET of the incident radiation spectrum. These simulation results suggest that data from solid walled detectors may be sufficient for the purpose of characterizing the radiation quality.

The values of yF for 10, 1, and 0.1 micrometer diameter sites are 162, 121, and 44 keV/µm respectively. This represents a 25% and 64% decreases in yF for each decrease in d by a factor of 10. The results suggest that the 10 micrometer diameter site may be larger than needed (at least when using solid walled detectors) to characterize radiations up to several GeV/nucleon. Reduction in the overall range of site sizes will make implementation of this approach for an operational detector much easier. The primary obstacle to producing an operational instrument is the large difference in the count rates in the different detectors, which is proportional to 1/r2. If, as now appears likely, detectors 2.5, 0.5, and 0.1 micrometers in diameter can be used, the ratios of count rates will be 25 and 625, values that are acceptable for experimental measurements at NASA Space Radiation Laboratory (NSRL). For an operational instrument, those ratios can be reduced by using a large number of the smaller detectors. We have developed a method for constructing large numbers of small detectors in order to reduce the size and weight of detectors used in neutron dosimetry. It should be possible to miniaturize this detector array approach to produce large numbers of 3 mm diameter detectors suitable for simulating 0.1 micrometer diameter sites.

Although it appears that solid walled detectors will be the right approach for designing an operational multi-site dosimetry system, work on simulating energy deposition in a homogeneous medium is continuing. We are hopeful that wall-less detectors will provide enough information about the spectrum of charged particles that such a system will be useful in evaluating the biological effectiveness of the mixture of primary particles and fragmentation products that occurs when shielding materials are evaluated experimentally at NSRL.

The Monte Carlo simulations have progressed to the point where it is now feasible to begin construction of an experimental detector system for use in confirming the simulation results with accelerator beams at NSRL. The current design calls for a metal sealed, stainless steel vacuum chamber approximately 15 cm in diameter and 20 cm high. This chamber will be equipped with a mounting system for three proportional counters, 0.3, 1.5, and 7.5 cm in diameter. These detectors, like those currently used on the space shuttle, will be cylindrical with field tubes defining the length of the active volume so that it is equal to the diameter. This configuration has the same mean chord length as a sphere of the same diameter. Electronics will be installed so that high and low gain spectra from each detector can be acquired simultaneously.

The results of Monte Carlo simulations of the probability density of lineal energy in solid walled detectors suggest that there is sufficient difference in the mean values as a function of site size to make it possible to evaluate the average velocity of the incident charged particles, as well as to determine the lineal energy spectrum and absorbed dose, using a detector consisting of three solid walled proportional counter cavities in a single detector chamber. It appears, from

	initial results, that simulated site sizes of 2.5, 0.5, and 0.1 micrometers diameter provide sufficient difference to characterize 1000 MeV/nucleon ions. Lower energy ions yet to be simulated by Monte Carlo methods, are expected to show a smaller ratio in yF values for 2.5 and 0. 5 diameter sites, relative to the ratio for 0.5 and 0.1 sites. This suggests that an operational detector system, suitable for use in space, is practical. The primary challenge will be to devise a detector system that will provide adequate count rate for all three simulated site sizes, and maintain the low noise characteristics required to adequately measure the frequency of the small events produced by delta rays. Simulations for wall less detectors (in process) will show even larger differences between yF values, which may prove to be very useful in evaluating radiation quality behind shielding materials in laboratory experiments. However, detectors utilizing wall less proportional counters are, of necessity, several times larger than detectors utilizing solid walled detectors, and may prove to be unnecessary for field measurements.
Bibliography Type:	Description: (Last Updated: 01/21/2010)
Articles in Peer-reviewed Journals	Wang X, Liu H, Xia Z, Braby LA. "Lineal Energy as a Function of Site Size for HZE Radiation." Radiation Protection Dosimetry, in press 2007. , Jan-2007